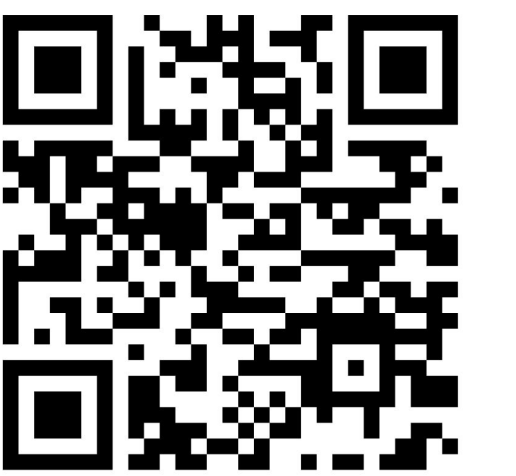
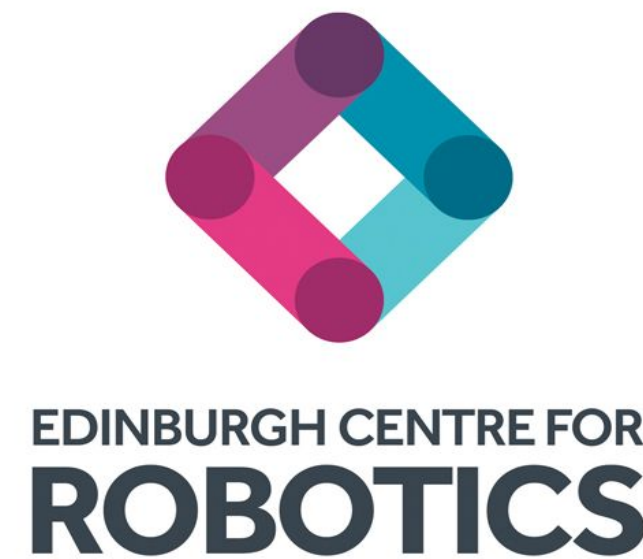


Tethered Multi-Robot Systems in Marine Environments

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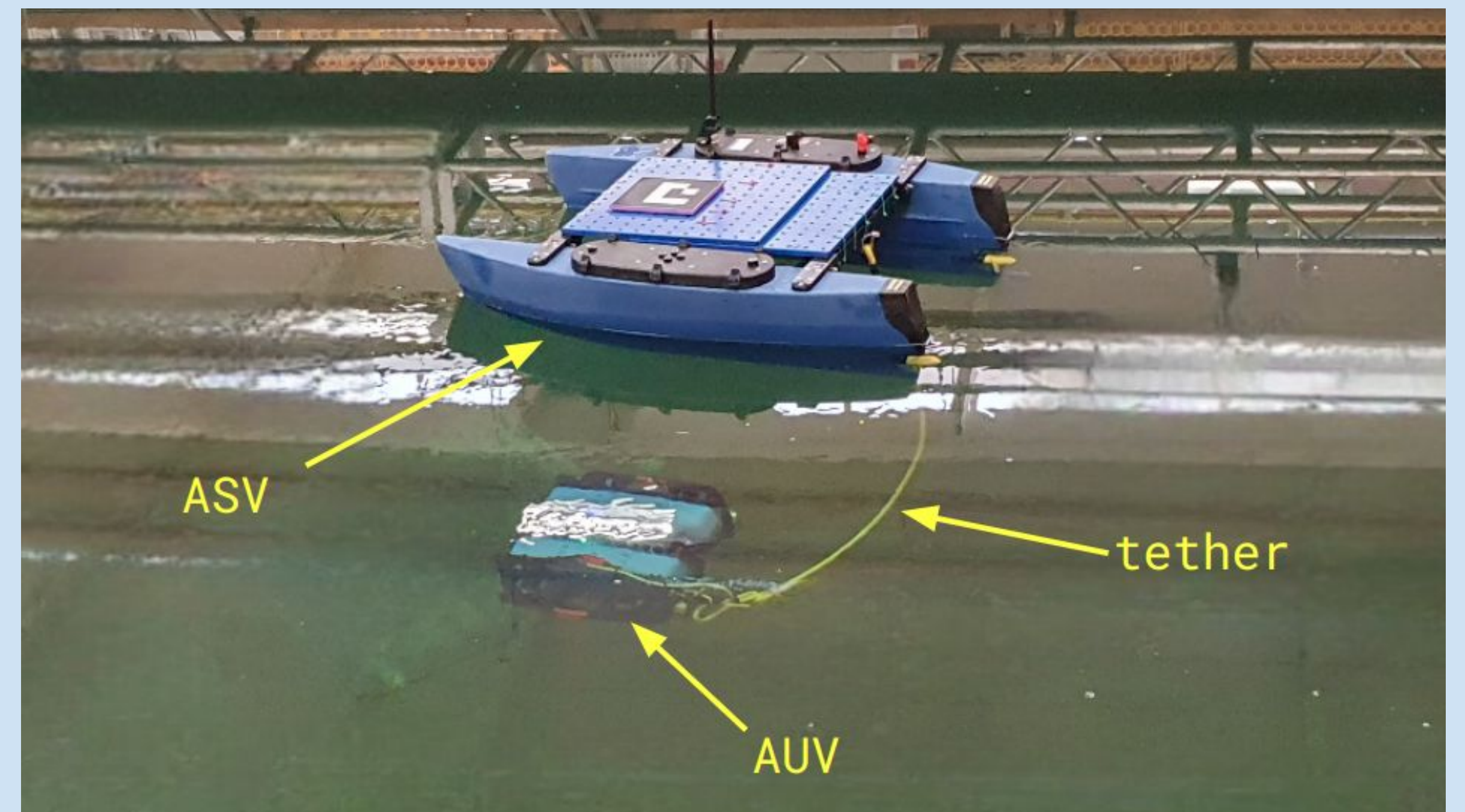
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Simulation Framework

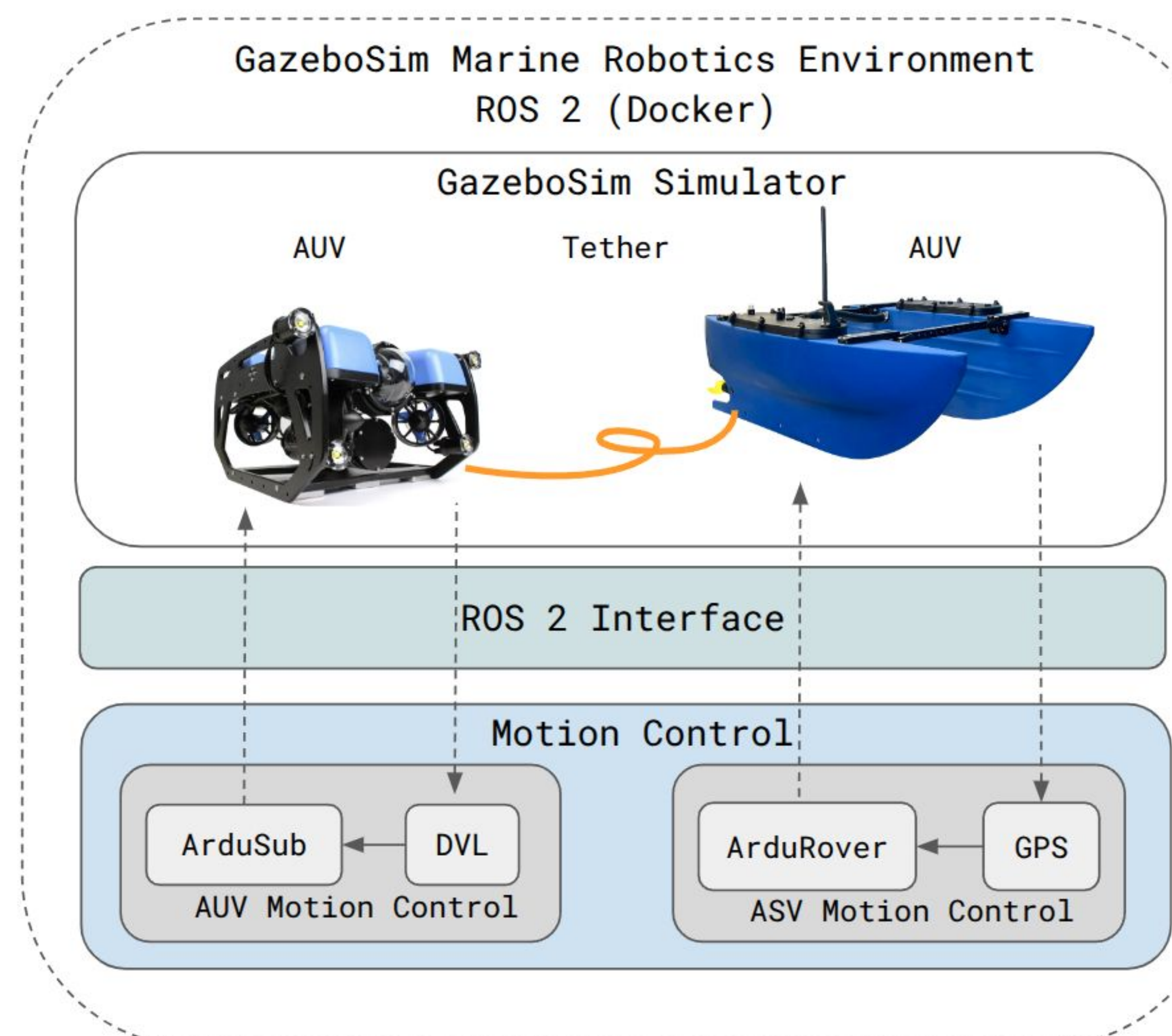
Goal

Develop a simulation framework to test motion control in **tethered** Autonomous Underwater Vehicle (**AUV**) and an Autonomous Surface Vehicle (**ASV**) systems operating in dynamic marine environments.



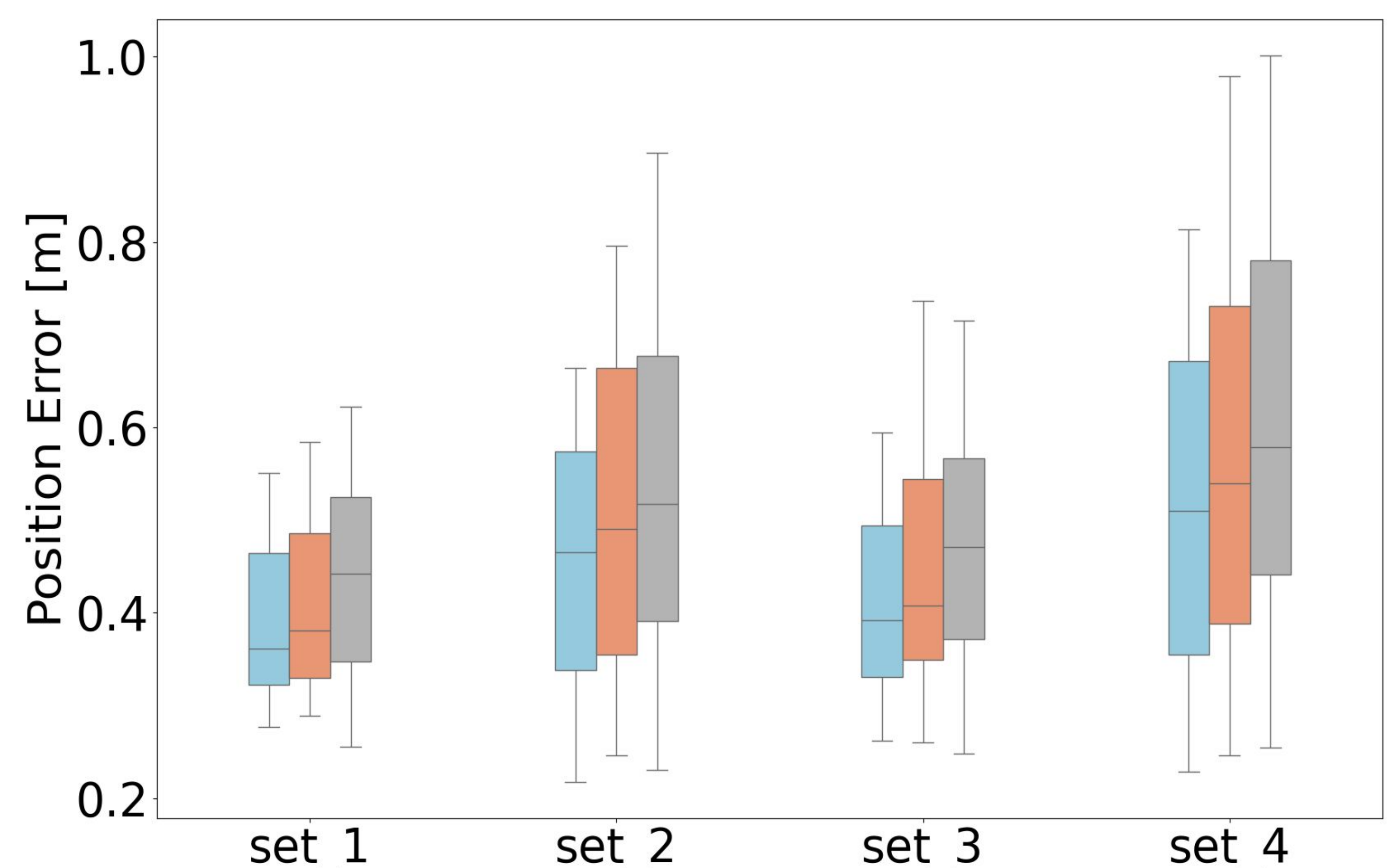
Architecture

The system uses a ROS 2-based simulation inside a Docker container, integrating an ASV and AUV in GazeboSim with realistic marine conditions like waves and currents. Both vehicles run ArduPilot (ArduSub and ArduRover) via SITL, allowing realistic control and sensor feedback through MAVLink.



Test and Results

We tested the system under moderate and rough sea conditions using different tether lengths and control setups. Results showed that disturbance-aware methods improved accuracy, and longer tethers increased positioning errors due to higher environmental impact.



Errors for ASV and AUV at different tether lengths: 5m (blue), 10m (orange), and 15m (grey) under rough sea conditions.